TIØ4145 Corporate Finance Chapters 9 and 10

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What is a share?

What determines the price of a company's shares?

Are you able to predict how the share price will develop?

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Balance sheet

Total Assets

- Fixed (non-current)
- Current

Liabilities and Equity

- Equity
- Long-term debt
- Short-term debt

Retained earnings

Dividends

- Revenue
- - Costs
- = Gross profit
- - Tax
- = Net profit

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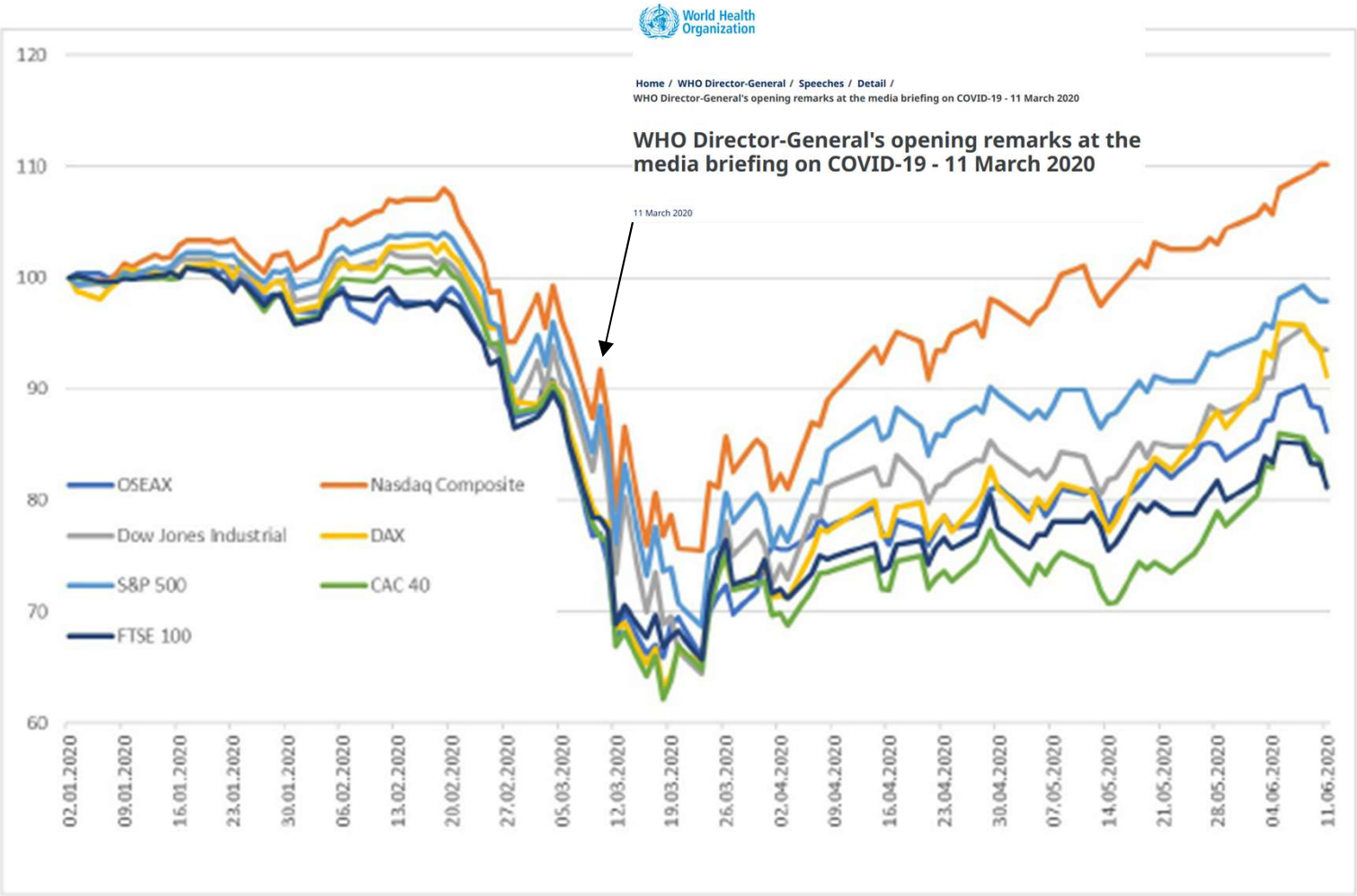

Figure 1: Relative performance of major indices and the Oslo Stock Exchange All Index (OSEAX) in the first months of 2020.

Størdal, S. et al. (2020) Norwegian stock market behaviour during the initial phase of the COVID-19 pandemic. Beta - Scandinavian Journal of Business Research. 34 (2): 207-221. https://doi.org/10.18261/issn.1504-3134-2020-02-04

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What are return of a share?

- Capital gains
- Dividends

The value of an equity is £100. The company earns £100 extra in cash. What is the investor's portfolio at t=1?

100% Dividends	0% Dividends	50% Dividends
• Price = £100 • Dividends = £100 • Total = £200	• Price = £200 • Dividends = £0 • Total = £200	• Price = £150 • Dividends = £50 • Total = £200

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Three Common Dividend Patterns

- Zero Growth
- Constant Growth
- Differential Growth

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Chapter 9

- 9.1 The Dividend-Discount Model
- 9.2 Applying the Dividend-Discount Model
- 9.3 Total Payout (self-study) and Free Cash Flow Valuation Models
- 9.4 Valuation Based on Comparable Firms
- 9.5 Information, Competition, and Stock Prices

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Valuing Stocks

- Chapter 9 learning objectives
  - You are able to use the **dividend-discount model** to compute the value of a dividend-paying company's stock, whether the dividends grow at a constant rate starting now or at some time in the future.
  - You can calculate the total **return of a stock**, given the dividend payment, the current price, and the previous price.
  - You can discuss **the determinants of future dividends** and their growth
  - Given the retention rate and the return on new investment, you can calculate the **growth rate** in dividends, **earnings**, and **share price**.
  - You can describe why **cutting** the firm's **dividend** may **raise** the **stock price**.
  - Assuming a firm has a long-term **constant growth** rate after time N + 1, you calculate the terminal value of the stock at time N.
  - You know how to use the **discounted free cash flow model** to calculate the value of stocks.
  - You can use comparable firm **multiples** to estimate a stock's value.
  - You can explain why **several valuation models** are required to value a stock.

Example: One-Year Investor

- A One-Year Investor
  - Potential Cash Flows:
    - Dividend
    - Sale of Stock
- Timeline for One-Year Investor

- Since the cash flows are risky, we must discount them at the equity cost of capital.

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Dividend-Discount Model

$$P_0 = \left(\frac{Div_1 + P_1}{1 + r_E} \right)$$

What is r_E?

- If the current stock price were less than this amount, expect investors to rush in and buy it, driving up the stock's price.
- If the stock price exceeded this amount, selling it would cause the stock price to quickly fall.
- Total Return:

$$r_E = \frac{Div_1 + P_1}{P_0} - 1 = \underbrace{\frac{Div_1}{P_0}}_{\text{Dividend Yield}} + \underbrace{\frac{P_1 - P_0}{P_0}}_{\text{Capital Gain Rate}}$$

- *The expected total return of the stock should equal the expected return of other investments available in the market with equivalent risk.*

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Dividend-Discount Model (cont.)

- What is the price if we plan on holding the stock for N years?

$$P_0 = \frac{Div_1}{1 + r_E} + \frac{Div_2}{(1 + r_E)^2} + \cdots + \frac{Div_N}{(1 + r_E)^N} + \frac{P_N}{(1 + r_E)^N}$$

  - This is known as the **Dividend Discount Model.**

- Note that the above equation holds for any horizon *N*. Thus all investors (with the same beliefs) will attach the same value to the stock, independent of their investment horizons.

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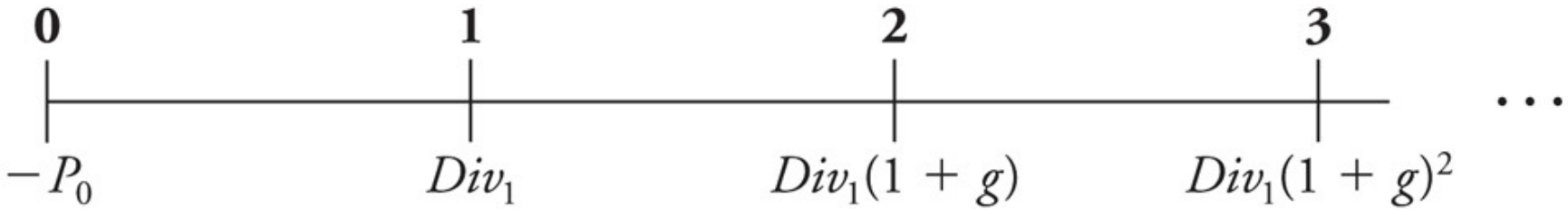
Applying the Dividend-Discount Model

- **Constant Dividend Growth:** The simplest forecast for the firm's future dividend states that they will grow at a constant rate, *g*, forever.



- The value of the firm depends on the current dividend level, the cost of equity, and the growth rate.

$$P_0 = \frac{Div_1}{r_E - g}$$

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Example: Applying the Dividend-Discount Model

- Example: AT&T plans to pay $1.44 per share in dividends in the coming year. Its equity cost of capital is 8%. Dividends are expected to grow by 4% per year in the future. Estimate the value of AT&T's stock.

$$P_0 = \frac{Div_1}{r_E - g}$$

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What determines a growth rate in dividends?

- $Div_t = \frac{\text{Earnings}_t}{\text{Shares Outstanding}_t} \times \text{Dividend Payout Rate}_t$

- $\text{Change in Earnings} = \text{New Investment} \times \text{Return on New Investment}$
  $= \text{Earnings} \times \text{Retention Rate} \times \text{Return on New Investment}$

- $\text{Earnings Growth Rate} = \frac{\text{Change in Earnings}}{\text{Earnings}}$
  $= \frac{\text{Earnings} \times \text{Retention Rate} \times \text{Return on New Investment}}{\text{Earnings}}$
  $= \text{Retention Rate} \times \text{Return on New Investment}$

- If the dividend payout rate (and all other rates) is constant then the growth rate in dividends is equal to the growth rate of earnings.

$$g = \text{Retention Rate} \times \text{Return on New Investment}$$

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Dividends Versus Investment and Growth

- If a firm wants to increase its share price, should it *cut* its *dividend* and invest more, or should it *cut investment* and increase its dividend?
- The answer will depend on the profitability of the firm's investments.

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Example 9.3

Crane Sporting Goods expects to have earnings per share of $6 in the upcoming year. Rather than reinvest these earnings and grow, the firm plans to pay out all of its earnings as a dividend. With these expectations of no growth, Crane's current share price is $60. Suppose Crane could cut its dividend payout rate to 75% for the foreseeable future and use the retained earnings to open new stores. The return on its investment is these stores is expected to be 12%. Assuming its equity cost of capital is unchanged, what effect would this new policy have on Crane's stock price?

- $r_E = \frac{Div}{P_0} + g = 10\% + 0\% = 10\%$
- $g = \text{Retention rate } \times \text{Return on new investments} = 25\% \times 12\ \% = 3\%$
- $P_0 = \frac{Div}{r_E - g} = \frac{4.50}{0{,}1 - 0{,}03} = 64.29$
- Generally, however: *Cutting the firm's dividend to increase investment will raise the stock price if, and only if, the new investments have a positive NPV*.

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Changing growth rates

- We cannot use the constant dividend growth model to value a stock if the growth rate is not constant.
- For example, young firms often have very high initial earnings growth rates. During this period of high growth, these firms often retain 100% of their earnings to exploit profitable investment opportunities. As they mature, their growth slows. At some point, their earnings exceed their investment needs and they begin to pay dividends.

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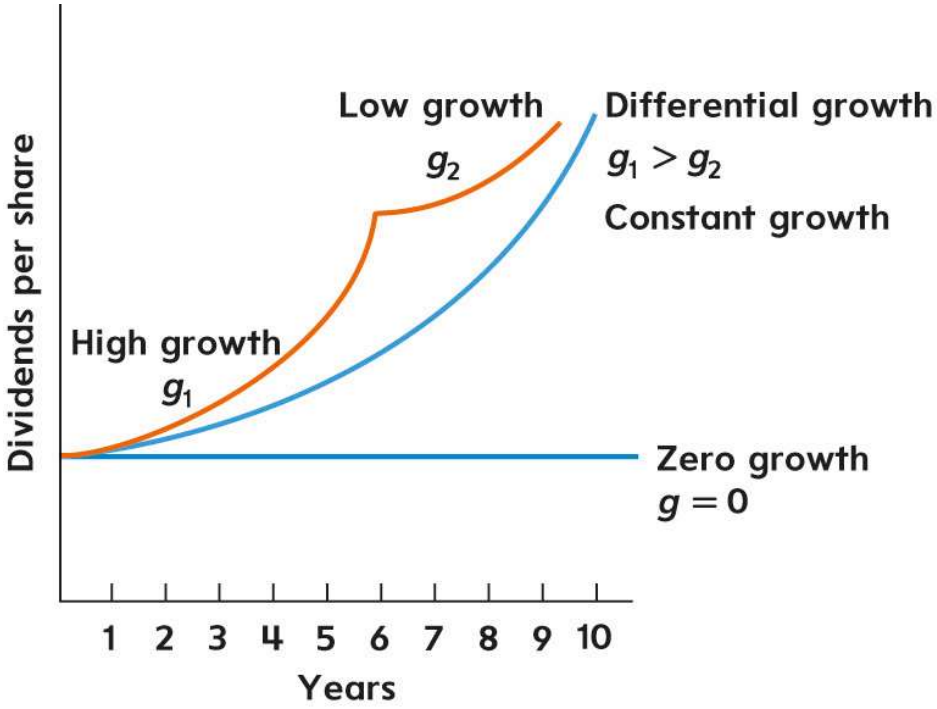
Three Common Dividend Patterns



Dividend growth models:

$$\text{Zero growth: } P_0 = \frac{Div_1}{R}$$

$$\text{Constant growth: } P_0 = \frac{Div_1}{R - g}$$

$$\text{Differential growth: } P_0 = \sum_{t=1}^{T} \frac{Div\,(1 + g_1)^t}{(1 + R)^t} + \frac{\dfrac{Div_{T+1}}{R - g_2}}{(1 + R)^T}$$

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Limitations of the Dividend-Discount Model

- There is a tremendous amount of uncertainty associated with forecasting a firm's dividend growth rate and future dividends.
- Small changes in the assumed dividend growth rate can lead to large changes in the estimated stock price.

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Some useful numbers

- **EBITDA**= Net Earnings + Tax + Interest + Depreciation/Appreciation
- **EBIT** = EBITDA – Depreciation/Appreciation
- **EBT** = EBIT – Interest
- Cash Flow (**CF**) = Cash from Operations
- Free Cash Flow (**FCF**) = CF – Cash from Investing
- Free Cash Flow to Equity (**FCFE**) = FCF – Cash from Financing
- Free Cash Flow to Firm (**FCFF**) = EBIT – Tax + Depreciation + Change in Net Working Capital – Capex {Also called Unlevered Free Cash Flow)

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Total Payout and Free Cash Flow Valuation Models

- Total Payout model for self study!
- Free Cash Flow Valuation Model
  - Determines the value of the firm to all investors, including both equity and debt holders

$$\text{Enterprise value} = \text{Market Value of Equity} + \text{Debt} - \text{Cash}$$

  - The enterprise value can be interpreted as the net cost of acquiring the firm's equity, taking its cash, paying off all debt, and owning the unlevered business.
  - Valuing the Enterprise

$$\text{Free Cash Flow} = \overbrace{EBIT(1-\tau_c)}^{\text{Unlevered net income}} + \text{Depreciation} - \text{Capital Expenditures} - \text{Increase in Net Working Capital}$$

Discounted Free Cash Flow Model

- Free Cash Flow is the Cash flow available to pay both debt holders and equity holders
- Discounted Free Cash Flow Model:
  - $V_0 = PV(\text{Future Free Cash Flow of Firm})$
  - $P_0 = \frac{\text{V}_0 + \text{Cash}_0 - \text{Deb}_0}{\text{Shares Outstanding}_0}$
- When implementing the Model we are discounting cash flows to both equity holders and debt holders:
  - Therefore the free cash flows should be discounted at the firm's weighted average cost of capital, r_{wacc}.
- If the firm has no debt, $r_{wacc} = r_E$.

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Discounted Free Cash Flow Model for N years

- Often, the terminal value is estimated by assuming no or a constant long-run growth rate g_{FCF} for free cash flows beyond year N, so
- $V_0 = \frac{FCF_1}{1+r_{wacc}} + \frac{FCF_2}{(1+r_{wacc})^2} + \cdots + \frac{FCF_N}{(1+r_{wacc})^N} + \frac{V_N}{(1+r_{wacc})^N}$
- $V_N = \frac{FCF_{N+1}}{r_{wacc}-g_{FCF}} = \frac{1+g_{FCF}}{r_{wacc}-g_{FCF}} FCF_N$

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Connection to capital budgeting

- The firm's free cash flow is equal to the sum of the free cash flows from the firm's current and future investments, so we can interpret the firm's enterprise value as the total NPV that the firm will earn from continuing its existing projects and initiating new ones.
- The NPV of any individual project represents its contribution to the firm's enterprise value. To maximize the firm's share price, we should accept projects that have a positive NPV.

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Valuation Based on Comparable Firms

- Estimate the value of the firm based on the value of other, comparable firms or investments that we expect will generate very similar cash flows in the future.
- Valuation Multiple: A ratio of firm's value to some measure of the firm's scale or cash flow
- The Price-Earnings Ratio (P/E Ratio): Share price divided by earnings per share
  - $\text{Forward}\,\frac{\mathrm{P}}{\mathrm{E}} = \frac{P_0}{EPS_1} = \frac{P}{E} = \frac{Div_1/EPS_1}{r_E - g} = \frac{\text{Dividend Payout Rate}}{r_E - g}$
- Firms with high growth rates, and which generate cash well in excess of their investment needs so that they can maintain high payout rates, should have high P/E multiples.

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Example: Valuation Based on Comparable Firms

- Example
  - (Also study textbook example 9.9): Best Buy Co. Inc. (BBY) has earnings per share of $2.22. The average P/E of comparable companies' stocks is 19.7. Estimate a value for Best Buy using the P/E as a valuation multiple.
- Other Multiples
  - Enterprise Value Multiples
    - $\frac{V_0}{EBITDA_1} = \frac{FCF_1/EBITDA_1}{r_{wacc} - g}$
  - Multiple of Sales
  - Price to book value of equity per share
  - Enterprise value per subscriber

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Common Valuation Multiples

Price multiples	Enterprise Value (EV) Multiples	Other ratios
Price-to-Earnings (P/E) Price-to-Book Value of equity (P/B) Price-to-Cash Flow (P/CF) Price-to-Sales (P/S)	EV/EBITDA EV/Operating profit	Dividend yield PEG-ratio (PE/Growth)

Enterprise Value (EV) = Market value of equity + Debt – Cash and Cash Equivalents

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Valuation Based on Comparable Firms (cont.)

- Limitations of Multiples
  - When valuing a firm using multiples, there is no clear guidance about how to adjust for differences in expected future growth rates, risk, or differences in accounting policies.
  - Comparables only provide information regarding the value of a firm relative to other firms in the comparison set.
  - Using multiples will not help us determine if an entire industry is overvalued.
- Comparison with Discounted Cash Flow Methods
  - Discounted cash flows methods have the advantage that they can incorporate specific information about the firm's cost of capital or future growth.
  - The discounted cash flow methods have the potential to be more accurate than the use of a valuation multiple.

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Stock Prices and Multiples for the Footwear Industry, January 2006

Ticker	Name	Stock Price ($)	Market Capitalization ($ millions)	Enterprise Value ($ millions)	P/E	Price/ Book	Enterprise Value/ Sales	Enterprise Value/ EBITDA
KCP	Kenneth Cole Productions	26.75	562	465	16.21	2.22	0.90	8.36
NKE	NIKE, Inc.	84.20	21,830	20,518	16.64	3.59	1.43	8.75
PMMAY	Puma AG	312.05	5,088	4,593	14.99	5.02	2.19	9.02
RBK	Reebok International	58.72	3,514	3,451	14.91	2.41	0.90	8.58
WWW	Wolverine World Wide	22.10	1,257	1,253	17.42	2.71	1.20	9.53
BWS	Brown Shoe Company	43.36	800	1,019	22.62	1.91	0.47	9.09
SKX	Skechers U.S.A.	17.09	683	614	17.63	2.02	0.62	6.88
SRR	Stride Rite Corp.	13.70	497	524	20.72	1.87	0.89	9.28
DECK	Deckers Outdoor Corp.	30.05	373	367	13.32	2.29	1.48	7.44
WEYS	Weyco Group	19.90	230	226	11.97	1.75	1.06	6.66
RCKY	Rocky Shoes & Boots	19.96	106	232	8.66	1.12	0.92	7.55
DFZ	R.G. Barry Corp.	6.83	68	92	9.20	8.11	0.87	10.75
BOOT	LaCrosse Footwear	10.40	62	75	12.09	1.28	0.76	8.30
				Average (excl. KCP)	**15.01**	**2.84**	**1.06**	**8.49**
				Max (relative to Ave.)	+51%	+186%	+106%	+27%
				Min (relative to Ave.)	−42%	−61%	−56%	−22%

Stock Valuation Techniques: The Final Word

- No single technique provides a final answer regarding a stock's true value. All approaches require assumptions or forecasts that are too uncertain to provide a definitive assessment of the firm's value.
- Most real-world practitioners use a combination of these approaches and gain confidence if the results are consistent across a variety of methods.

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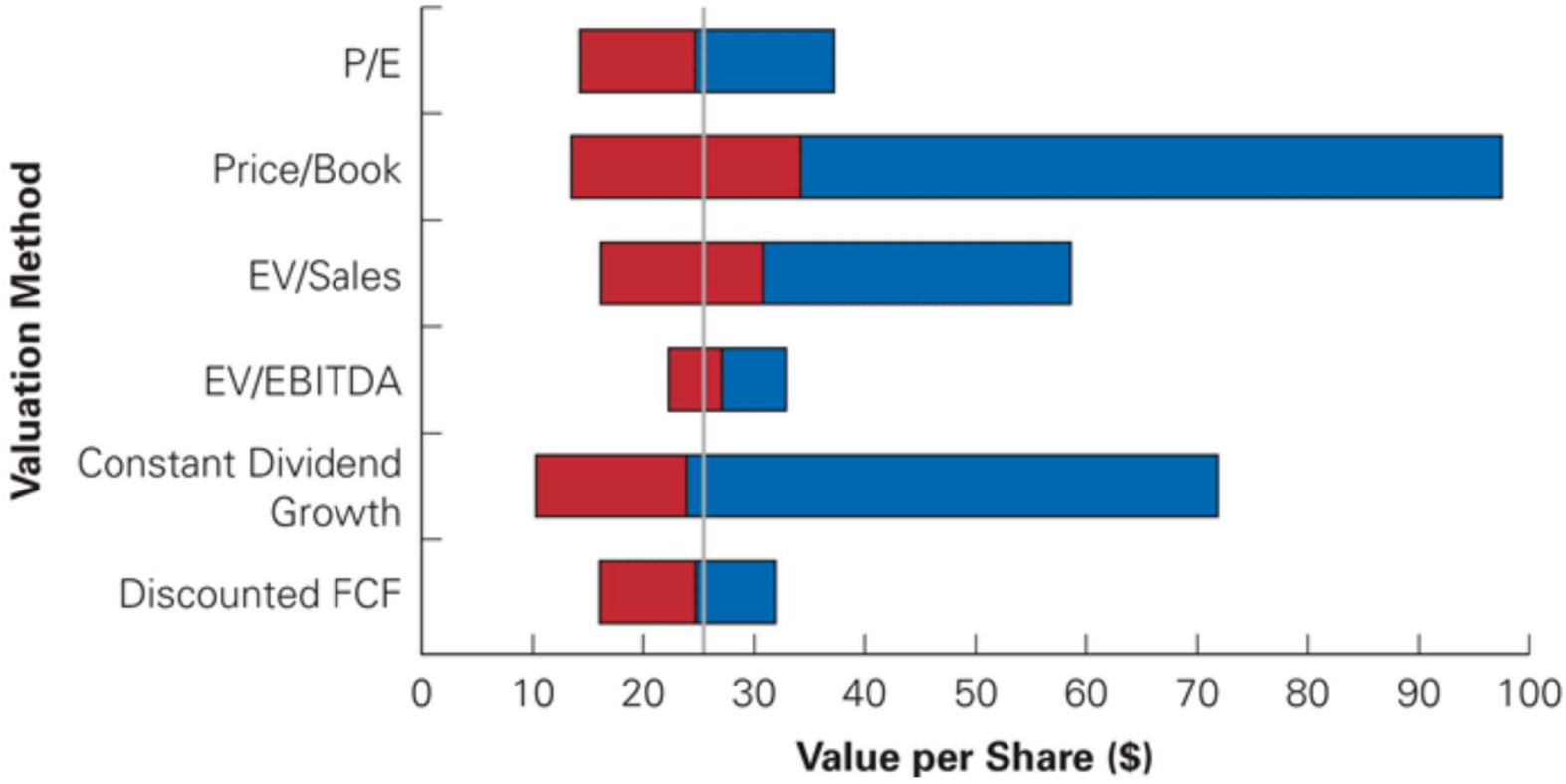
Range of Valuations for KCP Stock Using Alternative Valuation Methods



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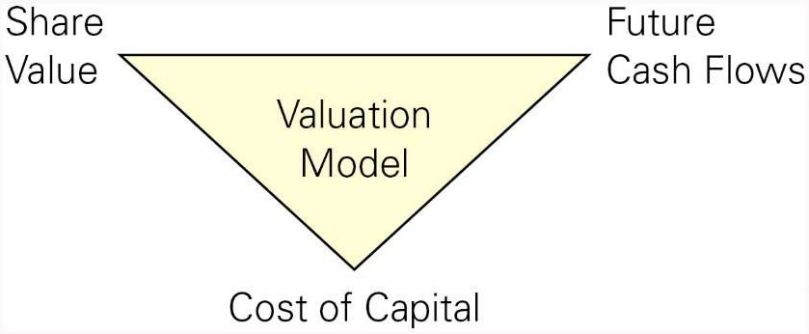
Information, Competition, and Stock Prices

- Our valuation model links the firm's future cash flows, its cost of capital, and its share price
- Given accurate information about any two of these variables, a valuation model allows us to make inferences about the third variable



- For a publicly traded firm, its current stock price should already provide very accurate information, aggregated from a multitude of investors, regarding the true value of its shares
  - Based on its current stock price, a valuation model will tell us something about the firm's future cash flows or cost of capital

Competition and Efficient Markets

- Efficient Markets Hypothesis
  - Implies that securities will be fairly priced, based on their future cash flows, given all information that is available to investors.
- Public, Easily Interpretable Information
  - In this situation, we expect the stock price to react nearly instantaneously to such news
- Private or Difficult-to-Interpret Information
  - Private information will be held by a relatively small number of investors
  - In this case, the efficient markets hypothesis will not hold in the strict sense
    - However, as these informed traders begin to trade, they will tend to move prices, so over time prices will begin to reflect their information as well
    - If the profit opportunities from having private information are large, others will devote the resources needed to acquire it
    - In the long run, we should expect that the degree of “inefficiency” in the market will be limited by the costs of obtaining the private information

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Efficient markets hypothesis more formally

- Fama* specifies contents of available information set in 3 categories:
  - weak form market efficiency: all past price histories are fully reflected in current prices (technical analysis is inefficient)
  - semi-strong form of market efficiency: prices fully reflect all publicly available information (fundamental analysis is inefficient)
  - strong form efficiency: all information reflected in current prices, incl. private and inside information.

Categories overlap, strong form efficiency implies semi-strong form whichimplies weak form efficiency.

**Eugene Fama, 2013 Nobel laureate in economics, widely recognized as the "father of modern finance."*

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Textbook Example 9.13

Phenyx Pharmaceuticals has just announced the development of a new drug for which the company is seeking approval from the Food and Drug Administration (FDA). If approved, the future profits from the new drug will increase Phenyx's market value by $750 million, or $15 per share given its 50 million shares outstanding. If the development of this drug was a surprise to investors, and if the average likelihood of FDA approval is 10%, what do you expect will happen to Phenyx's stock price when this news is announced? What may happen to the stock price over time?

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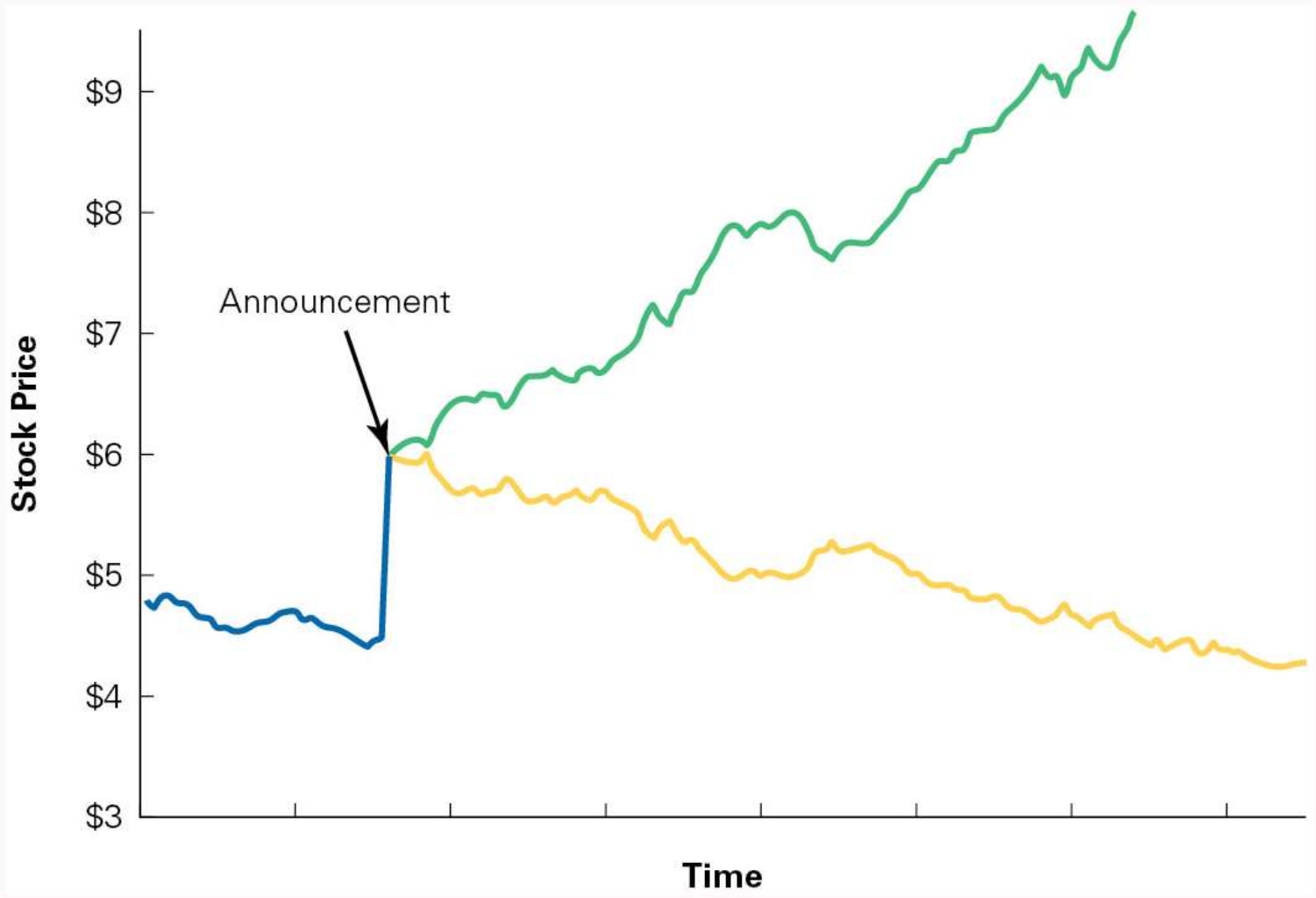
Figure 9.4 Possible Stock Price Paths for Example 9.13



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The Efficient Markets Hypothesis Versus No Arbitrage

- The efficient markets hypothesis states that securities with **equivalent risk** should have the **same expected return**
- An arbitrage opportunity is a situation in which two securities with **identical cash flows** have **different prices**

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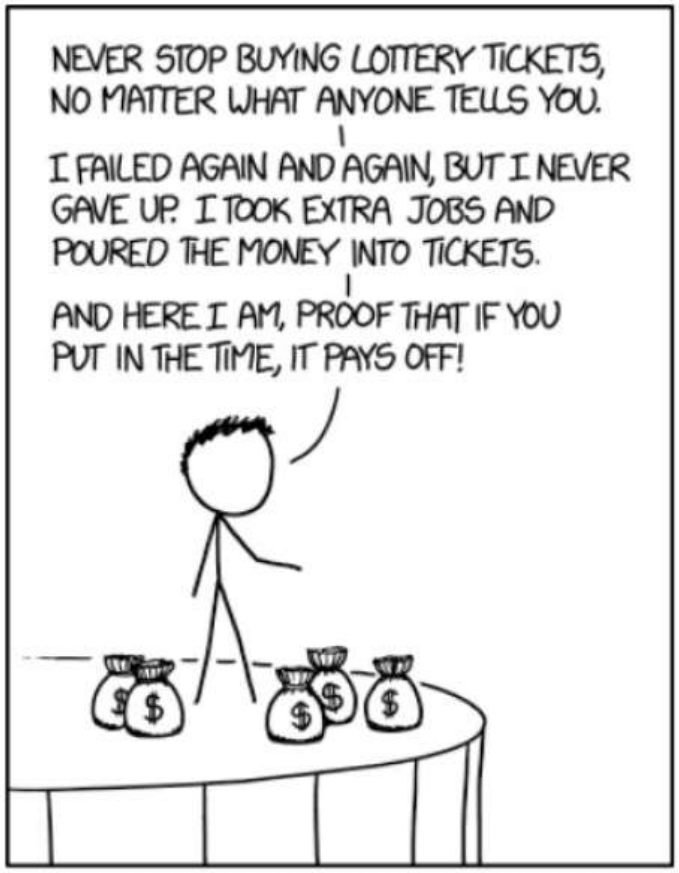
Markets offer no free lunch

- Does *not* say that
  - people cannot get lucky
  - people cannot repeatedly get lucky
- Luck determined by probability laws, not by investment skills



EVERY INSPIRATIONAL SPEECH BY SOMEONE SUCCESSFUL SHOULD HAVE TO START WITH A DISCLAIMER ABOUT SURVIVORSHIP BIAS.

Source https://xkcd.com/1827/

Capital Markets and the Pricing of Risk

- Chapter 10
  - 10.1 Risk and Return: Insights from 86 Years of Investor History
  - 10.2 Common Measures of Risk and Return
  - 10.3 Historical Returns of Stocks and Bonds
  - 10.4 The Historical Tradeoff Between Risk and Return
  - 10.5 Common Versus Independent Risk
  - 10.6 Diversification in Stock Portfolios
  - 10.7 Measuring Systematic Risk
  - 10.8 Beta and the Cost of Capital

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Chapter 10: Learning Objectives

- Develop a theory that explains the relationship between **average returns** and the **variability of returns**
- Define the **risk premium**
- Investigate the relationship between **historical data**
- Identify **different types of risks**, and their relation to the risk premium
- Determine the **cost of capital**

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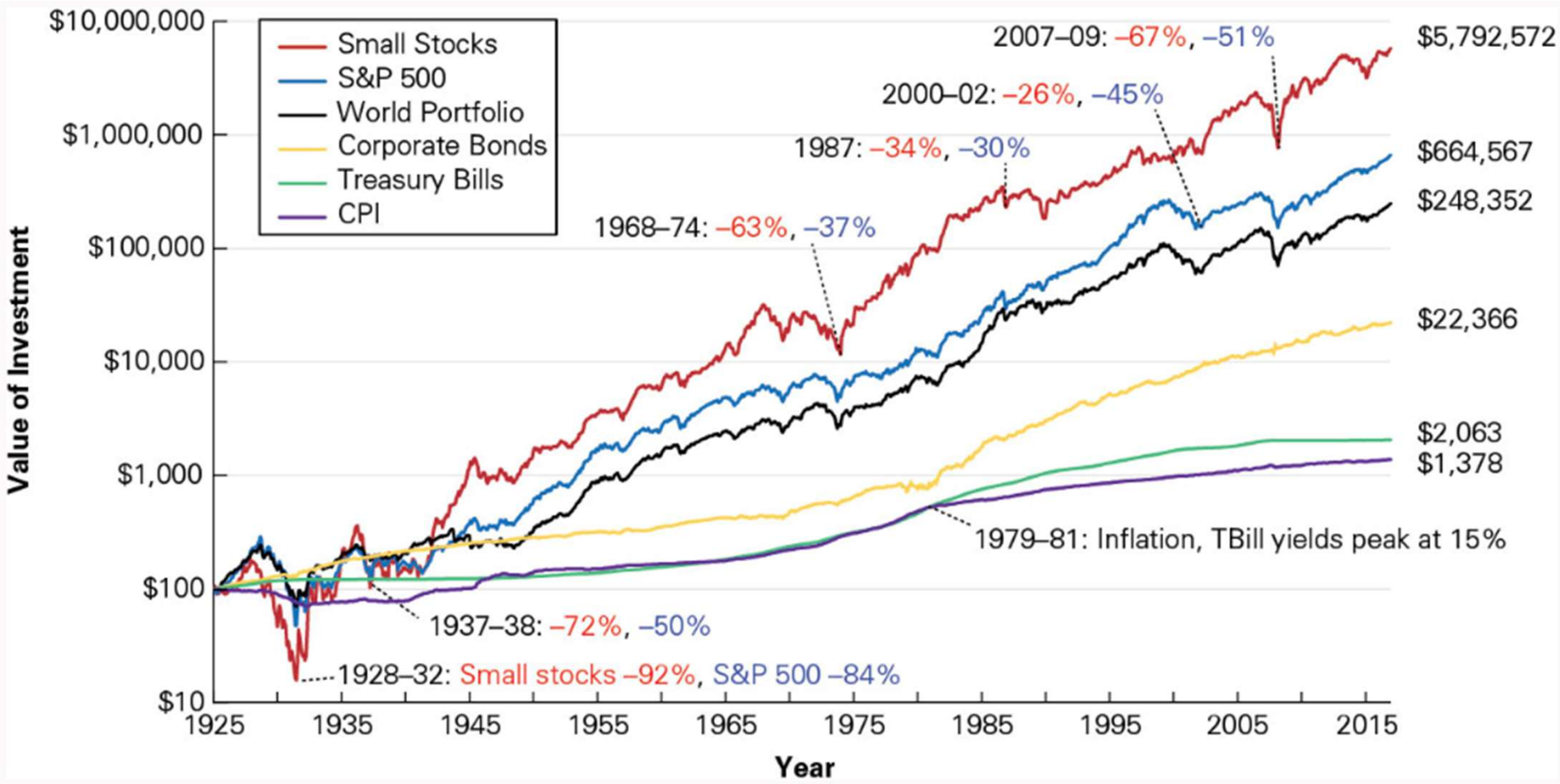
Insights from historical data



Value of $100 invested in 1925

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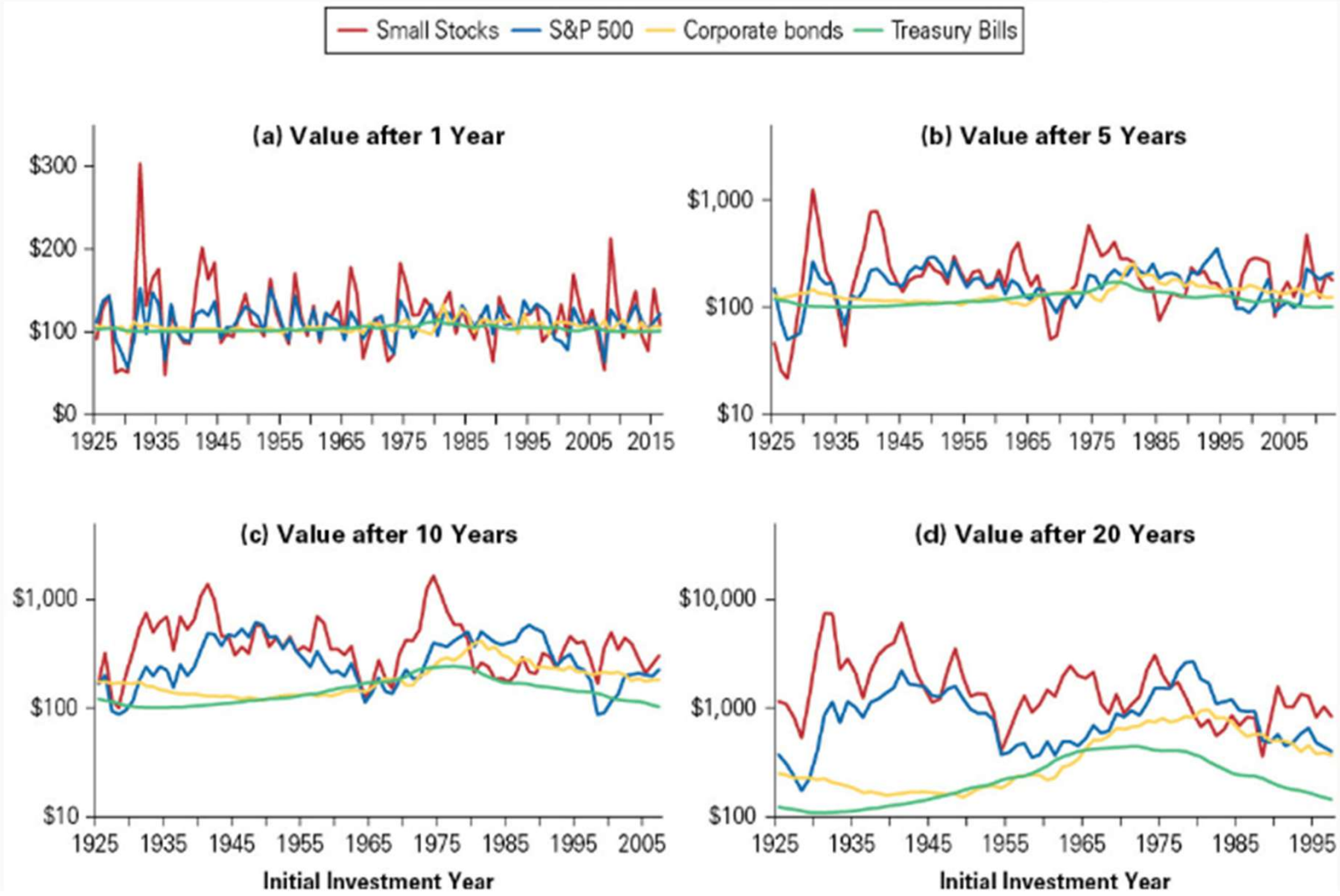
Shorter investment horizons



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What we know already

- Investors are averse to fluctuations in the value of their investments (Chapter 3).
- Therefore, they demand a risk premium to bear risk.
  - What is the relationship of risk and the risk premium investors demand?
  - First develop tools that allow to measure risk and return.

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Common Measures of Risk and Return

- Standard ways to define and measure risks
- Returns measure of the securities' performance
  - Percentage increase in the value of an investment per dollar initially invested in the security.
  - Example:

$$\textit{Total return} = \frac{P_1 + CF_1 - P_0}{P_0}$$

    - CF – cash flow payed by security
    - P – price

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Probability distributions

- When an investment is risky, there are different returns it may earn.
- Each possible return has some likelihood of occurring.
- This information is summarized with a probability distribution, which assigns a probability, p_R , that each possible return, R , will occur.

		Probability Distribution	
Current Stock Price ($)	Stock Price in One Year ($)	Return, R	Probability, P_R
	140	0.40	25%
100	110	0.10	50%
	80	−0.20	25%

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Expected return

- Expected Return (Mean Return)
  - Calculated as a weighted average of the possible returns, where the weights correspond to the probabilities.

$$\textit{Expected return} = E[R] = \sum_R p_R \times R$$

  - Expected return is the return we would earn on average if we could repeat the investment many times (drawing from the same distribution)

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Variance and Standard Deviation

- Variance
  - The expected squared deviation from the mean

$$Var[R] = E\left[(R - E[R])^2\right] = \sum_R p_R \times (R - E[R])^2$$

- Standard Deviation
  - The square root of the variance

$$SD[R] = \sqrt{Var[R]}$$

  - In finance, the standard deviation of a return is also referred to as its volatility.
  - Standard deviation is easier to interpret because it's in the same units as returns.

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Historical Returns of Stocks and Bonds

- In most situations we do not know the explicit probability distribution.
- How can we estimate and compare risk and return without this information?
- Popular approach: extrapolate from historical data
- How to compute average returns and volatilities using historical stock market data?

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Computing Historical Returns

- Realized Return
  - The return that actually occurs over a particular time period.

$$R_{t+1} = \frac{P_{t+1} + D_{t+1} - P_t}{P_t} = \frac{D_{t+1}}{P_t} + \frac{P_{t+1} - P_t}{P_t}$$

= Dividend Yield + Capital Gain Rate

  - If a stock pays dividends at the end of each quarter, with realized returns $R_{Q1}, \ldots, R_{Q4}$ each quarter, then its annual realized return, R_{annual}, is computed as:

$$1 + R_{annual} = (1 + R_{Q1})(1 + R_{Q2})(1 + R_{Q3})(1 + R_{Q4})$$

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Example

Date	Price	Dividend	Return
31/12/07	58.69		
31/01/08	64.44	0.26	
30/04/08	63.94	0.26	
31/07/08	48.50	0.26	
31/10/08	54.88	0.29	
31/12/08	53.31		

Date	Price	Dividend	Return
31/12/11	6.73		
31/03/12	5.72		
30/06/12	4.81		
30/09/12	5.20		
31/12/12	2.29		

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Example (cont.)

- What are the realized annual returns?

Date	Price	Dividend	Return
31/12/07	58.69		
31/01/08	61.44	0.26	
30/04/08	63.94	0.26	
31/07/08	48.50	0.26	
31/10/08	54.88	0.29	
31/12/08	53.31		

Date	Price	Dividend	Return
31/12/11	6.73		
31/03/12	5.72		
30/06/12	4.81		
30/09/12	5.20		
31/12/12	2.29		

- We compute each period's return. For example, the return from December 31, 2007 to January 31, 2008 is:

$$R_{t+1} = \frac{61.44 + 0.26 - 58.69}{58.69} = 5.13\%$$

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Example

- What are the realized annual returns?

Date	Price	Dividend	Return
31/12/07	58.69		
31/01/08	61.44	0.26	5.13%
30/04/08	63.94	0.26	4.49%
31/07/08	48.50	0.26	-23.74%
31/10/08	54.88	0.29	13.75%
31/12/08	53.31		-2.86%

Date	Price	Dividend	Return
31/12/11	6.73		
31/03/12	5.72		-15.01%
30/06/12	4.81		-15.91%
30/09/12	5.20		8.11%
31/12/12	2.29		-55.96%

- We compute each period's return. For example, the return from December 31, 2007 to January 31, 2008 is:

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30/09/12	5.20		8.11%
31/12/12	2.29		-55.96%

- We then calculate the annual returns

$$R_{2008} = (1.0513)(1.0449)(0.7626)(1.1375)(0.9714) - 1 = -7.43\%$$

$$R_{2012} = (0.8499)(0.8409)(1.0811)(0.4404) - 1 = -65.97\%$$

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Example

- What are the realized annual returns?

Date	Price	Dividend	Return
31/12/07	58.69		
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30/09/12	5.20		8.11%
31/12/12	2.29		-55.96%

- Note that, since the stock did not pay dividends during 2012, the return can also be computed as:

$$R_{2012} = \frac{2.29 - 6.73}{6.73} = -65.97\%$$

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Realized Return for the S&P 500, Microsoft, and Treasury Bills, 2005–2017

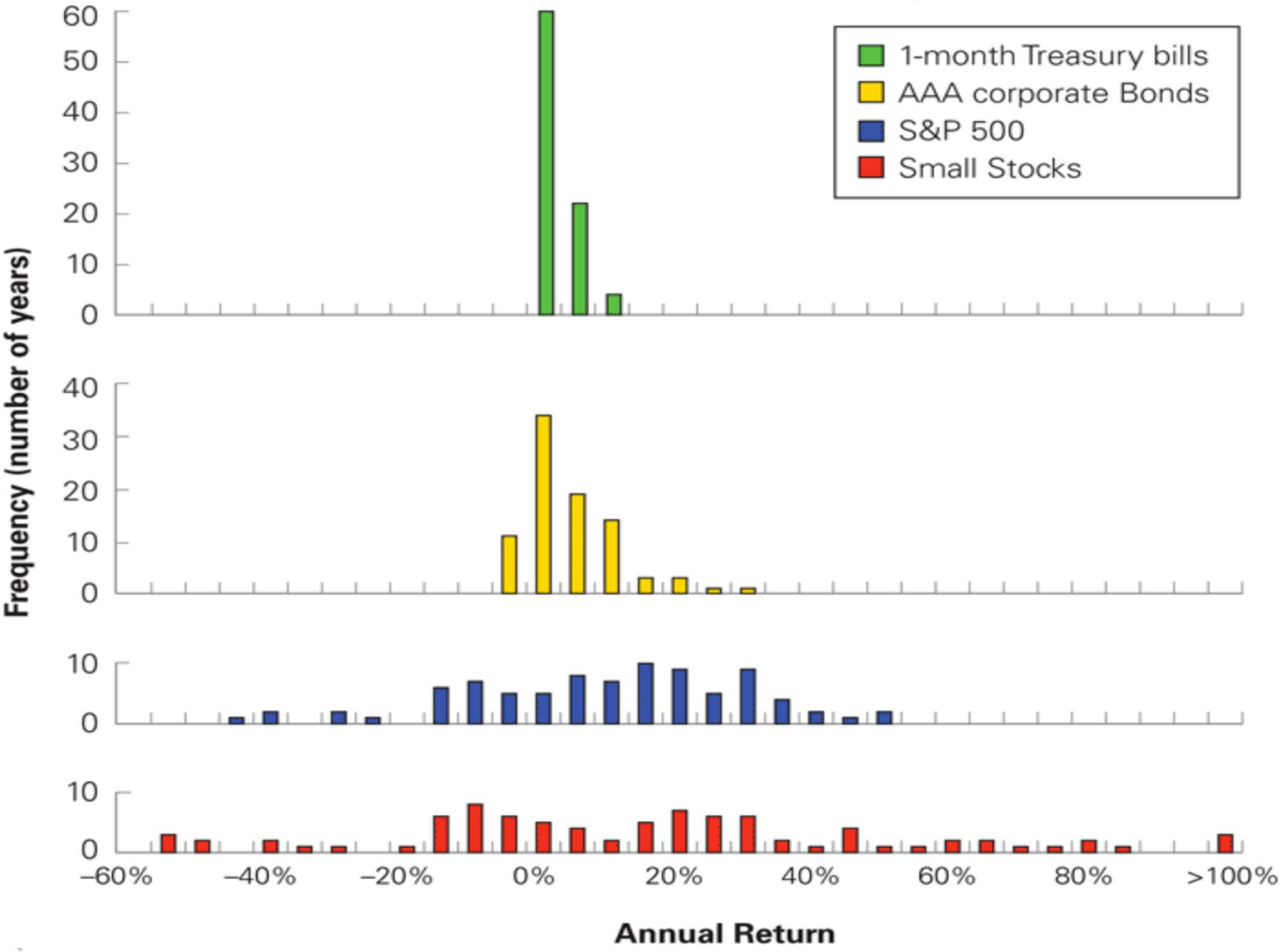
TABLE 10.2 Realized Return for the S&P 500, Microsoft, and Treasury Bills, 2002–2011

Year End	S&P 500 Index	Dividends Paid*	S&P 500 Realized Return	Microsoft Realized Return	1-Month T-Bill Return
2001	1148.08				
2002	879.82	14.53	−22.1%	−22.0%	1.6%
2003	1111.92	20.80	28.7%	6.8%	1.0%
2004	1211.92	20.98	10.9%	8.9%	1.2%
2005	1248.29	23.15	4.9%	−0.9%	3.0%
2006	1418.30	27.16	15.8%	15.8%	4.8%
2007	1468.36	27.86	5.5%	20.8%	4.7%
2008	903.25	21.85	−37.0%	−44.4%	1.5%
2009	1115.10	27.19	26.5%	60.5%	0.1%
2010	1257.64	25.44	15.1%	−6.5%	0.1%
2011	1257.60	26.59	2.1%	−4.5%	0.0%

*Total dividends paid by the 500 stocks in the portfolio, based on the number of shares of each stock in the index, adjusted until the end of the year, assuming they were reinvested when paid.

Source: Standard & Poor's, Microsoft and U.S. Treasury Data

Empirical Distribution of Annual Returns, 1926–2011



Average Annual Return

- Average Annual Return of a security
  - If R_t is the realized return of a security in year t, then the average annual return $\bar{R}$ for year 1 thourgh T is

$$\bar{R} = \frac{1}{T}(R_1 + R_t + \cdots + R_T) = \frac{1}{T}\sum_{t=1}^{T} R_t$$

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The Variance and Volatility of Returns

- Variance Estimate Using Realized Returns

$$Var[\bar{R}] = \frac{1}{T-1}\sum_{t=1}^{T}(R_t - \bar{R})^2$$

- The estimate of the standard deviation is the square root of the variance.

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Examples

- Average Annual Returns for U.S. Small Stocks, Large Stocks (S&P 500), Corporate Bonds, and Treasury Bills; 1926–2021

Investment	Average Annual Return
Small Stocks	18.6 %
S&P 500	12.2 %
Corporate bonds	6.2 %
Treasury bills	3.3 %

- Volatility of U.S. Small Stocks, Large Stocks (S&P 500), Corporate Bonds, and Treasury Bills, 1926–2021

Investment	Return Volatility (Standard Deviation)
Small Stocks	38.6 %
S&P 500	19.7 %
Corporate bonds	6.3 %
Treasury bills	3.1 %

Using Past Returns to Predict the Future

- Two difficulties with this approach:
  - We do not know what investors expected in the past; we can only observe the actual returns that were realized.
  - The average return is just an estimate of the true expected return, and is subject to estimation error.

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Standard Error

- A statistical measure of the degree of estimation error.
  - Standard Error of an Estimate of the Expected Return

$$SE^{*}(\text{Average of Independent, Identical Risks}) = \frac{SD(\text{Individual risks})}{\sqrt{\text{Number of Observations}}}$$

  - 95% confidence interval

$$\text{Historical Average return} \pm (2 \times \text{Standard Error})$$

    - For the <u>S&P500</u>: $12.2\ \% \pm 2\frac{19.7\%}{\sqrt{96}} = 12.2\% \pm 4.0\%$

* The book reads SD, but it's more common to denote the Standard Error as SE

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Different returns

- Arithmetic Average Returns vs Geometric (Compound) Annual returns
  - Arithmetic annual return: $\frac{1}{T}\sum_{i=1}^{T} R_i$
  - Geometric (compound) annual return: $[(1+R_1)\times(1+R_2)\times \ \dots \times(1+R_T)]^{\frac{1}{T}}-1=\left[\prod_{i=1}^{T}(1+R_i)\right]^{\frac{1}{T}}-1$
  - Continous compound (log) annual returns: $\frac{1}{T}\sum_{i=1}^{t} ln(1+R_i)$

Year	Price	Return
1	100	
2	50	-50%
3	100	+100%
Average arithmetic return		+25 %
Average geometric return		0 %

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The Historical Tradeoff Between Risk and Return

- The Returns of Large Portfolios
  - Excess Returns
    - The difference between the average return for an investment and the average return for T-Bills (as risk-free investment).

Investment	Return Volatility (Standard Deviation)	Excess Return Average Return in Excess of Treasury Bills)
Small Stocks	38.6 %	15.3 %
S&P 500	19.7 %	8.9 %
Corporate bonds	6.3 %	2.9 %
Treasury bills	3.1 %	0.0 %

Volatility Versus Excess Return, 1926–2021.

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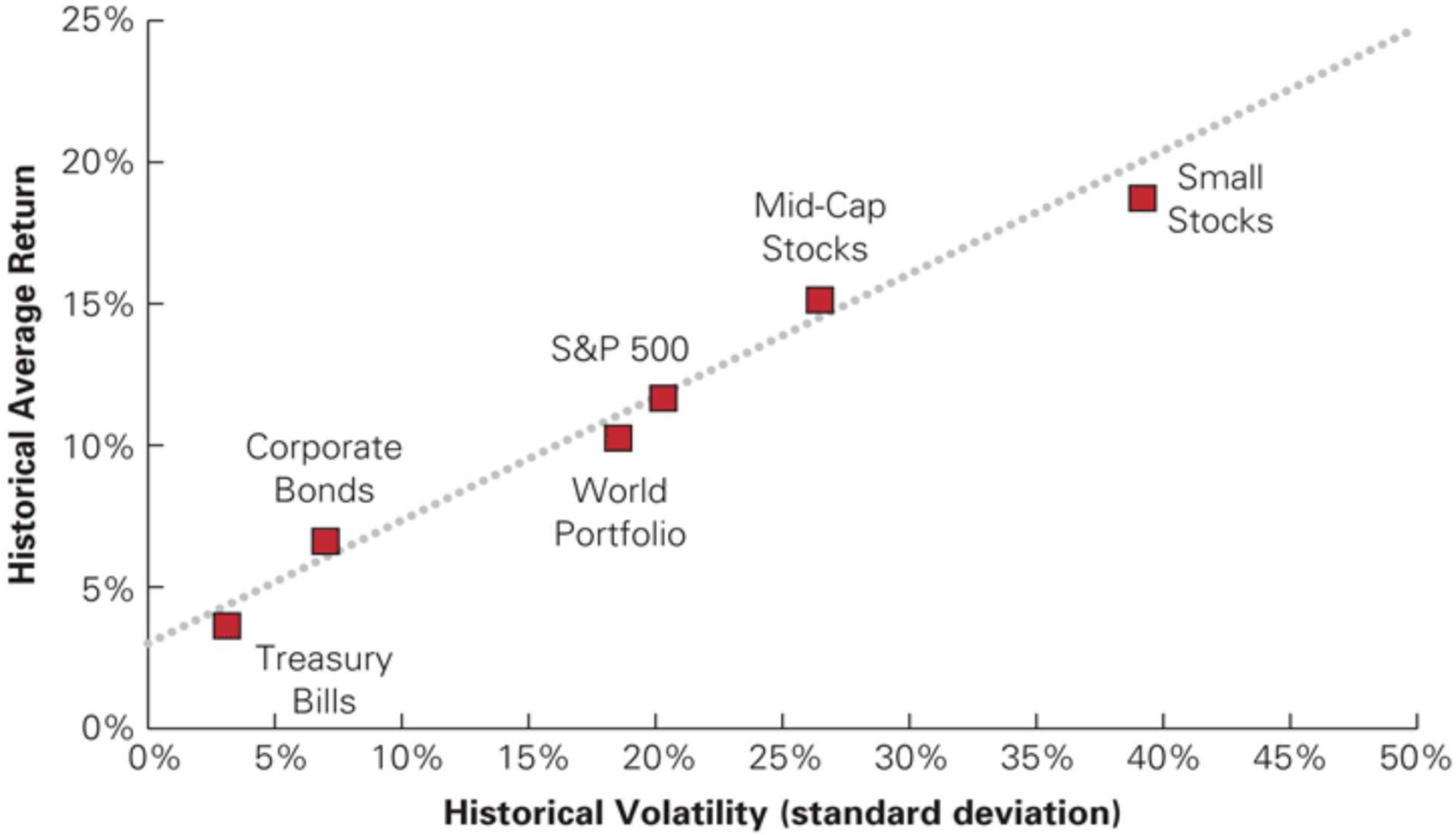
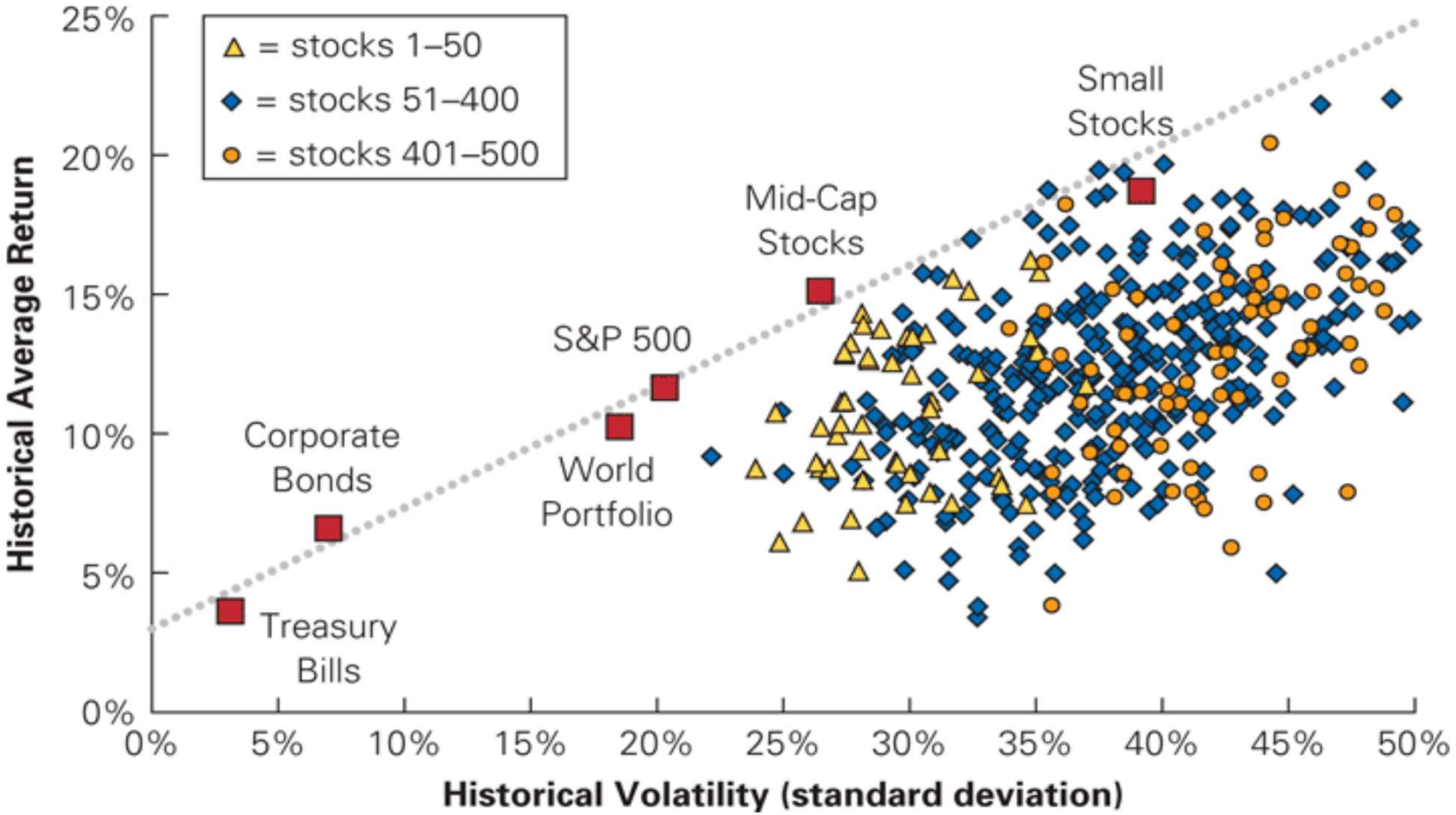
The Historical Tradeoff Between Risk and Return

- Large portfolios



The Historical Tradeoff Between Risk and Return

- Individual stocks, ranked by size



The Returns of Individual Stocks

- Is there a positive relationship between volatility and average returns for individual stocks?
  - There is no precise relationship between volatility and average return for individual stocks.

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Puzzles

- Two puzzles:
  - Why do all stocks tend to have higher risk and lower returns than large portfolios?
  - Why is there no clear relationship between volatility and return?

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Common Versus Independent Risk

- Common risk
  - Risk that is perfectly correlated. Risk that affects all securities.
- Independent Risk
  - Risk that is uncorrelated. Risk that affects a particular security.
- Diversification
  - The averaging out of independent risks in a large portfolio

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Diversification in Stock Portfolios

- Firm-Specific Versus Systematic Risk:
  - Firm Specific News
    - Good or bad news about an individual company
  - Market-Wide News
    - News that affects all stocks, such as news about the economy

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Types of Risks

- **Independent Risks**

Also known as:

- Firm-Specific Risk
- Idiosyncratic Risk
- Unique Risk
- Unsystematic Risk
- Diversifiable Risk

- **Common Risks**

Also known as:

- Systematic Risk
- Undiversifiable Risk
- Market Risk

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Example

- Which of the following risks of a stock are likely to be firm-specific, diversifiable risk, and which are likely to be systematic risks?
  a. The risk that the countries are going into lock down due to COVID-19 outbreak.
  b. The risk that a product design is faulty and the product must be recalled.
  c. The risk that the new product you expect your R&D division to produce will not materialize.
  d. The risk that your main production plant is shut down due to a tornado.

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Example: answer

- Which of the following risks of a stock are likely to be firm-specific, diversifiable risk?
  a. The risk that the countries are going into lock down due to COVID-19 outbreak.
  b. The risk that a product design is faulty and the product must be recalled.
  c. The risk that the new product you expect your R&D division to produce will not materialize.
  d. The risk that your main production plant is shut down due to a tornado.

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No Arbitrage and the Risk Premium

- If the diversifiable risk of stocks were compensated with an additional risk premium, then investors could buy the stocks, earn the additional premium, and simultaneously diversify and eliminate the risk.
- By doing so, investors could earn an additional premium without taking on additional risk.
- This opportunity to earn something for nothing would quickly be exploited and eliminated.

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No Arbitrage and the Risk Premium

- The risk premium for diversifiable risk is zero, so investors are not compensated for holding firm-specific risk.
  - Diversification does however not reduce systematic risk.
  - Because investors are risk-averse, they will demand a risk-premium to hold systematic risk.
- The risk premium of a security is determined by its systematic risk and does not depend on its diversifiable risk.

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Measuring Systematic Risk

- Identifying Systematic Risk
  - To measure the systematic risk of a stock, determine how much of the variability of its return is due to systematic risk versus unsystematic risk.
  - To determine how sensitive a stock is to systematic risk, look at the average change in the return for each 1% change in the return of a portfolio that fluctuates solely due to systematic risk.
  - 1st Step: Find a portfolio that contains only systematic risk.

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Measuring Systematic Risk

- Efficient Portfolio
  - A portfolio that contains only systematic risk. There is no way to reduce the volatility of the portfolio without lowering its expected return.
  - A key question in finance: Best way to identify an efficient portfolio?
- Market Portfolio
  - An efficient portfolio that contains all shares and securities in the market
    - The S&P 500 is often used as a proxy for the market portfolio.

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Sensitivity to Systematic Risk

- Beta (β)
  - The expected percent change in the excess return of a security for a 1% change in the excess return of the market portfolio.
    - Beta differs from volatility. Volatility measures total risk (systematic plus unsystematic risk), while beta is a measure of only systematic risk

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Betas with Respect to the S&P 500 for Individual Stocks (based on monthly data for 2007–2012)

Company	Ticker	Industry	Equity Beta
General Mills	GIS	Packaged Foods	0.20
Consolidated Edison	ED	Utilities	0.28
The Hershey Company	HSY	Packaged Foods	0.28
Abbott Laboratories	ABT	Pharmaceuticals	0.31
Newmont Mining	NEM	Gold	0.32
Wal-Mart Stores	WMT	Superstores	0.35
Clorox	CLX	Household Products	0.39
Kroger	KR	Food Retail	0.42
Altria Group	MO	Tobacco	0.43
Amgen	AMGN	Biotechnology	0.44
McDonald's	MCD	Restaurants	0.47
Procter & Gamble	PG	Household Products	0.47
Pepsico	PEP	Soft Drinks	0.51
Coca-Cola	KO	Soft Drinks	0.54
Johnson & Johnson	JNJ	Pharmaceuticals	0.59
PetSmart	PETM	Specialty Stores	0.75
Molson Coors Brewing	TAP	Brewers	0.78
Nike	NKE	Footwear	0.91
Microsoft	MSFT	Systems Software	1.01
Southwest Airlines	LUV	Airlines	1.09
Intel	INTC	Semiconductors	1.09
Whole Foods Market	WFM	Food Retail	1.10
Foot Locker	FL	Apparel Retail	1.11
Oracle	ORCL	Systems Software	1.12
Amazon.com	AMZN	Internet Retail	1.13
Google	GOOG	Internet Software and Services	1.14

Interpreting Beta

- A security's beta is related to how sensitive its underlying revenues and cash flows are to general economic conditions.
- Stocks in cyclical industries are likely to be more sensitive to systematic risk and have higher betas than stocks in less sensitive industries.

Beta and the Cost of Capital

- Estimating the Risk Premium
  - Market risk premium = $\mathrm{E}[R_{Mkt}] - r_f$
  - Reflects investor's risk tolerance and determines the market price of risk in the economy.
  - The market risk premium is the reward investors expect to earn for holding a portfolio with a beta of 1.
- Adjusting for Beta
  - Estimating a Traded Security's Cost of Capital of an investment from its Beta:

$$\begin{aligned}\mathrm{E}[R] &= \text{Risk Free Interest Rate} + \text{Risk Premium} \\ &= r_f + \beta(\mathrm{E}[R_{Mkt}] - r_f)\end{aligned}$$

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Capital Asset Pricing Model

- CAPM
  - $\mathrm{E}[R] = r_f + \beta(\mathrm{E}[R_{Mkt}] - r_f)$
  - It is the most important method for estimating the cost of capital that is used in practice.
- Problem
  - Assume the economy has a 60% chance of the market return will 15% next year and a 40% chance the market return will be 5% next year.
  - Assume the risk-free rate is 6%.
  - If Microsoft's beta is 1.18, what is its expected return next year?

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Summary

- Define a probability distribution, the mean, the variance, the standard deviation, and the volatility.
- Compute the realized or total return for an investment.
- Using the empirical distribution of realized returns, estimate expected return, variance, and standard deviation (or volatility) of returns.
- Use the standard error of the estimate to gauge the amount of estimation error in the average.
- Discuss the volatility and return characteristics of large stocks versus large stocks and bonds.
- Describe the relationship between volatility and return of individual stocks.

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Summary

- Define and contrast idiosyncratic and systematic risk and the risk premium required for taking each on.
- Define an efficient portfolio and a market portfolio.
- Discuss how beta can be used to measure the systematic risk of a security.
- Use the Capital Asset Pricing Model to calculate the expected return for a risky security.
- Use the Capital Asset Pricing Model to calculate the cost of capital for a particular project.
- Explain why in an efficient capital market the cost of capital depends on systematic risk rather than diversifiable risk.

- Next lecture: Portfolio Choice and the Capital Asset Pricing Model

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